MILITARY SPECIFICATION

HYDRAULIC JACK SYSTEM, ELEVATED CAUSEWAY; HYDRAULIC COMPONENTS AND SHELTER PROVISIONS

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a special application hydraulic jacking system complete with equipment storage shelters and auxiliary lift hardware for raising and lowering pontoon causeway sections.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-C-271 - Chains and Attachments, Welded and Weldless.
TT-E-522 - Enamel, Phenolic, Outside.
TT-P-664 - Primer, Coating, Synthetic, Rust-inhibiting, Lacquer-Resisting.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.

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MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-T-704 - Treatment and Painting of Material.
MIL-E-10062 - Engine, Preparation for Shipment and Storage of.
MIL-P-12742 - Primer Coating: Phenolic, Water Immersible.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification for.
MIL-C-18295 - Chain and Fittings for Fleet Moorings.
MIL-G-18458 - Grease, Wire Rope-Exposed Gear.
MIL-P-24441 - Paint, Epoxy-Polyamide, General Specification for.
MIL-S-28633 - Shelter, Equipment, Multipurpose.
MIL-I-45208 - Inspection System Requirements.
MIL-P-46093 - Primer Coating, Synthetic (For Brake Drums).

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - Colors.

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MIL-STD-129 - Marking for Shipment and Storage.
MS3367 - Strap, Tiedown, Electrical Components, Adjustable, Self Clinching, Plastic, Type I, Class I.
MS25036 - Terminal, Lug, Crimp Style, Copper, Insulated, Ring Tongue, Bell Mouthed, Type II, Class I (For 105 Deg C Total Conductor Temperature).

DRAWINGS

NAVAL CIVIL ENGINEERING LABORATORY (NCEL)

76-4-1F - Hydraulic Jack System Packaging and Tie Downs.
76-4-2F Rev. C - 50-Ton Chain Jack Canisters and Tie Down Details.
76-4-3F Rev. C - Chain Carts and Misc. (Rev.1).
76-4-4F Rev. B - Control Box Assembly.
76-4-5F Rev. C - Control Box Details.
76-4-6F Rev. B - Control Box Cam.
76-4-7F Rev. C - 50-Ton Chain Jack Assembly (with Automatic Control).
76-4-8F - Hydraulic Power Unit, Electrical Schematic.
76-4-9F Rev. B - Hydraulic Power Unit Assembly.
76-4-10 Rev. B - Hydraulic Power Unit Details.
76-4-11 Rev. A - Hydraulic Power Unit Details.
76-4-12 Rev. B - Hydraulic Power Unit Details.
76-4-13 Rev. B - Lift System Hydraulic Schematic and Hose Bundles.
76-4-14 - Hydraulic Power Unit Details.

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76-4-15 - DRAWING IS NOT PART OF THIS SPECIFICATION.
76-4-16 - DRAWING IS NOT PART OF THIS SPECIFICATION.
76-4-17 - Seabee Equipment Shelter Equipment Rails and Tie Downs.
76-4-18 Rev. A - 50-Ton Chain Jack Spanner Wrenches.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Structural Steel.
A307 - Carbon Steel Externally and Internally Threaded Standard Fasteners.
B209 - Sheet and and Plate, Aluminum Alloy.
B308 - Aluminum Alloy Standard Structural Shapes, Rolled or Extruded.
B429 - Structural Pipe and Tube, Aluminum Alloy Extruded.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC., (AWS)

Welding Inspection Manual.
D1.1 - Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W., 7th Street, Miami, FL 33125.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification, in conjunction with the Government-furnished drawings, covers the features, functions, arrangements, dimensions, performance, power, and details of the design of the components of a complete

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special application hydraulic jacking system. Jack system components consist of the following:

a. Diesel engine driven hydraulic power unit assembly(s).
b. 50-ton hydraulic jack(s).
c. Hose bundle(s).
d. Gimbal assembly(s).
e. Jack canister(s).
f. Chain cart(s).
g. Equipment shelter(s).
h. Spanner wrench(s).
i. Swivel hook(s).
j. Jaw-Jaw swivel(s).
k. Anchor shackle(s).

Equipment design criteria is controlled by Naval Civil Engineering Laboratory (NCEL) drawings.

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete component for first article inspection and testing as soon as practicable after award of contract or order. The first article may be either a preproduction model or an initial production item which conforms to all requirements of the specification. The first article shall be identical to the production items in accordance with the terms of the contract. The first article shall be inspected and tested by the contractor, in the presence of Government representatives, at the site selected by the contractor. Costs of testing including test scheduling, transportation, fuel, power, and material shall be borne by the contractor. The contractor is not responsible for travel expenses incurred by Government representatives. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with contract and specification requirements. No change or deviation from the approved first article will be acceptable without prior written approval of the contracting officer. The first article, after testing and approval, shall be retained by the contractor at his manufacturing facility for use by the Government inspector for comparative purposes in connection with the production units required to be delivered under the contract. In the event the contractor desires to deliver the test model as a contract item, it shall be delivered as the last item on the contract only after the contractor, at his own cost and expense, shall have completely cleaned, reconditioned, and overhauled the unit, making such replacements and modifications thereto as are required to make the unit acceptable as a contract item.

3.3 Inspection system requirements. Quality assurance inspection and tests necessary to substantiate product conformance, by the contractor, to contract requirements shall be in accordance with MIL-I-45208.

3.4 Drawings. Drawings forming a part of this specification are engineering design drawings. The contractor is responsible for preparing his own shop drawings. Where tolerances provided could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the items. No deviation from the prescribed dimensions or tolerances is permissible without authorization or prior approval by the contracting officer.

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3.4.1 Drawing review. The contractor is responsible for carefully and critically reviewing the contract drawings and specification prior to commencing construction. The contractor is responsible for correct performance of the contract despite any error, omission, discrepancy, or lack of clarity in the contract drawings or specification which should have been apparent as the result of such a review. If the contractor finds any such error, ommission, discrepancy, or lack of clarity, he shall immediately notify the Contracting Officer before commencing any work affected thereby, and present his solution for approval. However, if the contractor elects to perform work prior to such approval, he does so at his own risk.

3.4.2 Conflict. In the case of conflict between the contract drawings and this specification, this specification shall govern. In the case of conflict between this specification and other documents referenced herein, this specification shall govern. Silence of this specification relative to any details shown on the contract drawings or failure of the contract drawings to depict all details covered by this specification shall not be considered a conflict.

3.5 Responsibility for proper performance. The contractor shall not be relieved from fulfilling the full requirements of the contract drawings and specification, of the responsibility for producing satisfactory results, or of properly performing any work as a result of the following:

a. Absence of details where the essential features, functions, and arrangements are defined.
b. Mistakes in description of components or assembly details which, if not correct, would interfere with the proper performance of the items involved.

3.6 Interchangeability. All items with the same end usage shall be functionally and dimensionally interchangeable. This requirement includes parts, assemblies, and accessories.

3.7 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.7.1 Steel. Except where otherwise specified herein or in the design drawings, structural steel shall conform to ASTM A36.

3.7.2 Aluminum. Except where otherwise specified herein or in the design drawings, aluminum sheet and plate shall conform to ASTM B209; aluminum structural shapes shall conform to ASTM B308; aluminum structural pipe shall conform to ASTM B429; and aluminum alloy shall be 6061 and temper shall be T6.

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3.7.3 Fasteners. Except where otherwise specified herein or in the design drawings, steel fasteners shall be common hardware conforming to ASTM A307.

3.7.4 Hydraulic hose and fittings. All hydraulic hose and hydraulic hose fittings shall be as specified on the applicable design drawings. All low pressure (drain) hoses, not specified in the design drawings, shall be of suitable construction for their intended purpose. Inside diameter of hoses shall be as specified in the applicable design drawings. Where hydraulic hose is required and the inside diameter is not specified, in the design drawings, inside diameter size shall be selected in accordance with good commercial practice, but limited to the following nominal sizes: 1/4-, 3/8-, 1/2-, 3/4-, 1-, or 1-1/2 inch. Fittings not specified shall be of the reusable type and shall provide satisfactory flow/pressure characteristics for the item requirements.

3.7.5 Material identification. All materials, equipment, and machinery purchased, manufactured, or assembled, and all work performed under the contract shall be inspected by the contractor to verify conformance with specfication requirements prior to submission to the Government. Except as otherwise specified, the contractor may utilized his own facilites or any commercial laboratory acceptable to the Contracting Officer. The contractor shall offer to the Government only those supplies determined by the contractor to conform to the specification requirements. The Contracting Officer will reject defective or improper materials, equipment, machinery, workmanship.

3.7.6 Samples. The contractor shall, upon request of the Contracting Officer, furnish samples of materials and information concerning their quality and use. Wherever the identity or quality of an item is in doubt, it shall not be installed until its identity or quality has been positively established by tests conducted by the contractor.

3.8 Construction. Fabrications shall be constructed and assembled as indicated on the design drawings. Screw threads for threaded parts, not specified on the design drawings, shall conform to applicable sections of FED-STD-H28. Welding shall be in accordance with 3.14.2. Construction shall be such that welding operations shall be performed in a flat horizontal manner wherever possible, and shall be performed by a qualified welder (see 3.14.2.1). Where welding is performed in an alternate position, the welder must be qualified in the alternate position (see 3.14.2.1).

3.8.1 Hydraulic system. The hydraulic power unit assembly hydraulic system provides a means for operating the hydraulic jacks and other suitable hydraulically powered tools. The system shall have a maximum working pressure (relief valve setting) of 3,000 pounds per square inch (psi). The hydraulic power unit shall be capable of providing a minimum hydraulic oil flow rate of 30 gallons per minute (gpm) at not less than 3,000 psi. All system components shall be capable of withstanding a proof pressure that is equal to or greater than 1-1/2 times the system maximum working pressure without damage, permanent deformation, or external leakage. All hydraulic system components shall have a minimum burst pressure that is equal to or greater than three times the system relief valve setting. The system shall perform all functions at maximum design rates with a pressure that is less than the relief valve reseat pressure. The hydraulic system shall be cleaned free of all foreign matter (including weld slag and spatter, filings, cloth fibers, abrasives, etc.) before filling system with hydraulic fluid.

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2. APPLICABLE DOCUMENTS

3.8.2 Hydraulic hose assembly and installation. Hydraulic hose assemblies shall be provided in accordance with this specification and applicable NCEL drawings. Hydraulic hose requiring skiving shall be skived in accordance with the hose and fitting manufacturer's recommendations. Before attaching hose fittings to skived hose ends, the exposed reinforcing wire shall be sealed to prevent corrosion from salt water and high moisture environments. All installed hydraulic hoses shall be routed in accordance with good commercial practice to provide: shortest line, minimum number of bends, ease of maintenance and accessibility, and maximum protection. Provisions shall be made to prevent damage to installed hoses due to chafing. The hose installation shall cause no harmful stress concentration on the hose fittings. Hoses shall be supported, when necessary, to reduce excessive sagging at the fitting.

3.9 Components. Components shall be assembled, arranged, constructed, fabricated, modified, and stowed as specified herein and on the applicable NCEL drawings. Quantities of each jack system component shall be as specified (see 6.2).

3.9.1 Hydraulic power unit assembly. The hydraulic power unit assembly shall be constructed in accordance with the requirements of this specification and the following NCEL Drawings:

76-4-8F - Hydraulic Power Unit, Electrical Schematic.
76-4-9F - Hydraulic Power Unit Assembly.
76-4-10F - Hydraulic Power Unit Details.
76-4-11F - Hydraulic Power Unit Details.
76-4-12F - Hydraulic Power Unit Details.
76-4-13F - Lift System, Hydraulic Schematic and Hose Bundles.
76-4-14F - Hydraulic Power Unit Details.

In addition, the hydraulic power unit control panel shall be equipped with permanent labels or suitable plates to identify manually operated controls, pressure gages, meters, manually operated switches, manually operated buttons, manually operated electrical disconnects, etc. and to indicate possible hazardous conditions. The hydralic hose quick-disconnect panel and hydraulic control valve shall be color coded to match the jack idler wheels as per note #3 of NCEL Drawing 76-4-7F.

3.9.1.1 Engine. The hydraulic power unit shall be equipped with a Detroit Diesel, Model No. 1023-5100 (series 2-71) engine (manufactured by Detroit Diesel, Allison Division, General Motors Corporation) with a rated power output of 68 brake horsepower at 2,300 revolutions per minute. The engine shall be equipped to meet the requirements of this specification and the NCEL drawings, and shall be furnished with the following special equipment and accessories:

a. 12-volt direct current electric starter.
b. 12-volt alternator.
c. Air inlet housing.
d. 18-inch 6-blade right hand suction fan.
e. Crankshaft pulley.
f. Fuel and lubrication system filters.
g. Flywheel SAE#2 RKFD CLA 2174-023 CLU.
h. Fuel injectors - Model N60.

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i. Front and rear engine mounts - option No. 6Q43.
j. Intake air cleaner - option No. 6M38.
k. Fan guard and fan shroud - option No. 6K1C6.
l. Water connections - option No. 6K2A24.
m. Variable speed governor - option No. 6N4428.
n. Radiator and radiator shell - option No. 13A77.
o. Exhaust manifold, center outlet vertical - option No. 6L40.
p. Tachometer drive assembly.

(1) 1 - 5136249 adaptor, tachometer drive.
(2) 1 - 186647 bolt, 1/4x1 National Coarse thread.
(3) 1 - 103319 lockwasher, 1/4 inch.
(4) 1 - 5135935 gasket, tachometer drive adaptor.
(5) 2 - 186625 bolt, 5/16x7/8 National Coarse thread.
(6) 2 - 103320 lockwasher, 5/16 inch.
(7) 1 - 5145557 cable, flexible, tachometer drive.
(8) 1 - 5102049 adaptor, tachometer, 0.191, ratio 1.0:0.5.
(9) 1 - 5136370 adaptor, tachometer drive, cable.

3.9.1.2 Storage battery. The hydraulic power unit engine storage battery shall be furnished dry charged and sealed to prevent intrusion of atmospheric moisture.

3.9.1.3 Electrical system. Electrical system shall be constructed as specified herein and as specified on NCEL Drawing No. 76-4-8F. No. 10 and No. 14 gage insulated electrical wire shall conform to MIL-W-16878. Plastic, self clinching, tiedown, straps shall conform to MS3367 and crimp style lug terminals shall conform to MS25036. Heavy cable shall be provided and installed as specified on NCEL Drawing No. 76-4-8F.

3.9.2 50-ton hydraulic jack assembly. The 50-ton hydraulic jack assembly shall be part No. 24490 as produced by HYDRANAUTICS Inc., 6338 Lindmar Drive, P.O. Box 1068, Goleta, CA 93017, and shall be modified to conform to NCEL Drawing No. 76-4-7F. Means shall be provided to prevent a hydraulic pressure build-up between the wiper ring (sheet 7, part 1) and the rod seal (sheet 7, part 2). Jack chain idler wheels shall be painted (color coded) differently for each jack of each lift system as follows:

1 Jack - orange, No. 12246.
1 Jack - yellow, No. 13538.
1 Jack - blue, No. 15180.
1 Jack - red, No. 11105.
1 Jack - black, No. 17038. (Spare jack.)

Color chip numbers are in accordance with FED-STD-595.

3.9.3 Jack chain. Jack chain shall be stud link, 1-1/4 inch by 44 feet (nominal), 107 links long conforming to MIL-C-18295, group 1, half shot, with end links the same lay. One and one-fourth inch chain which conforms to MIL-C-18295 has a proof test load of 92,910 pounds (lb) and a break test load of 130,000 lb. The center link and every 12th link in both directions from the center link shall be painted yellow, leaving 11 unpainted links between each painted link and five unpainted links on each end. Primer coat for painted links shall conform to MIL-P-12742. Finish paint shall be lusterless enamel conforming to TT-E-522, type 1, yellow, color No. 13538 of FED-STD-595.

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3.9.4 Automatic hydraulic control box assembly. The automatic hydraulic control box assembly shall be constructed and arranged in accordance with the requirements of this specification and NCEL Drawing Nos. 76-4-4F, 76-4-5F, and 76-4-6F. The complete automatic hydraulic control box assembly shall be installed on the chain jack assembly in accordance with NCEL Drawing No. 76-4-7F. The automatic hydraulic control box assembly shall maintain a controlled hydraulic oil flow rate of 5 gpm at maximum hydraulic system pressure to the jack assembly.

3.9.4.1 Control box cam. The control box cam shall be constructed in accordance with the requirements of this specification and NCEL Drawing No. 76-4-6F.

3.9.5 Hose bundles. Jack hose bundles and power takeoff hose bundles shall be constructed in accordance with the requirements of this specification and NCEL Drawing No. 76-4-13F.

3.9.6 Gimbal assembly. The gimbal assembly shall be part No. 22264B as produced by HYDRANAUTICS Inc., 6338 Lindmar Drive, P.O. Box 1068, Goleta, CA 93017, and furnished with two additional pivot pins and five additional detent pins.

3.9.7 Jack canister. The jack canister shall be constructed in accordance with the requirements of this specification and NCEL Drawing No. 76-4-2F. The jack canister shall be compatible with the equipment shelter track system (see 3.9.8). All metal surfaces, except threaded connections, shall be finished in accordance with 3.10.

3.9.8 Chain cart assembly. The chain cart assembly shall be constructed in accordance with the requirements of this specification and NCEL Drawing No. 76-4-3F. The chain cart assembly shall be compatible with the equipment shelter track system (see 3.9.8). All metal surfaces, except threaded connections, shall be finished in accordance with 3.10.

3.9.9 Equipment shelters. Equipment shelters which meet the requirements of MIL-S-28633, shall be modified in accordance the requirements of this specification and in accordance with NCEL Drawing No. 76-4-17F. When specified (see 6.2), equipment shelters which meet the requirements of MIL-S-28633 shall be furnished by the Government for modification to the contractor. The equipment rails and tie down system shall be constructed and furnished in accordance with the requirements of this specification and NCEL Drawing No. 76-4-17F. Paint shall be applied to all metallic surfaces in accordance with 3.10 through 3.10.2.2 of this specification. Paint shall not be applied in accordance with the paint notes of NCEL Drawing No. 76-4-17F.

3.9.10 Spanner wrench(s). Spanner wrench(s) shall be constructed in accordance with the requirements of this specification and NCEL Drawing No. 76-4-18F. The piston spanner wrench and the upper gland spanner wrench shall be compatible with the hydraulic jack assembly.

3.9.11 Swivel hook(s). Swivel hook(s) shall be Crosby-Laughlin, Part No. 322A, having a rated safe working load of 11 tons, equipped with an S-4055 latch assembly attached. Hook and latch assembly shall be modified to provide lanyard operation of the latch.

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3.9.12 Jaw-jaw swivel(s). Jaw-jaw swivel(s) shall be Johnson Block Co., Model No. 19-JJ, having a rated safe working load of 19 tons.

3.9.13 Anchor shackle(s). Anchor shackle(s) shall be 1-3/8 inch, with a screw pin, and shall conform to RR-C-271. In addition, minimum breaking strength shall be not less than 99,800 lb; maximum breaking strength shall be not greater than 120,000 lb. Shackle(s) shall be zinc-coated.

3.10 Cleaning and painting. Cleaning and painting of assemblies shall be as specified herein. Paint shall be applied to all surfaces normally painted in accordance with standard commercial practice. Rubber, chrome, and glass are examples of surfaces which are not normally painted in standard commercial practice. Except for the diesel engine, surfaces of prepainted purchased parts and equipment shall be painted in accordance with 3.10 through 3.10.2.2 of this specification.

3.10.1 Cleaning. All surfaces of structural steel fabrications shall be cleaned by blasting. Blast cleaning shall be to a near white metal blast or cleaner. Near white metal blast is described as a surface with 95 percent of each square inch free of all visible residue with the remaining 5 percent of each square inch having no more than very light shadows or streaks caused by rust stains or mill scale oxides. Blast cleaning shall utilize either dry sand or crushed grit made of cast iron or steel or synthetic grit. Grit shall be of a maximum particle size no larger than that passing a 16 mesh screen. All aluminum surfaces shall be cleaned in accordance with MIL-T-704. Cleaned surfaces shall be primed as soon as possible, but in no case shall the time between cleaning and application of the primer coat exceed 4 hours. Epoxy-polyamide paint shall be applied in accordance with 3.10.2.

3.10.2 Epoxy-polyamide painting. Epoxy-polyamide paint shall conform to MIL-P-24441. MIL-P-24441 formulations shall be applied only to dry surfaces in an atmosphere having relative humidity not greater than 80 percent and a temperature of 70° Fahrenheit (F) or higher. When ambient atmosphere does not meet this requirement, all painting shall be accomplished in a controlled humidity environment. There shall, as a minimum, be not less than 16 hours of curing between each coat. If more than 7 days elapse between epoxy coats, the surface shall be cleaned with water and detergent, rinsed with fresh water, dried, and than a tack coat (1 to 2 mils wet film thickness) of the last coat applied before application of the next full coat called for. The two components of all the MIL-P-24441 coatings should be mixed in equal volume by first thoroughly stirring each component separately and then thoroughly mixing them together. After mixing, there should be a waiting period - about 2 hours at 50° to 60°F, 1 to 1-1/2 hours at 60° to 70°F, and 1/2 to 1 hour above 70°F - before applying the coating to insure complete curing later. If spray or roller painting is utilized, all corners and welds shall first be thoroughly brush coated before each coat of paint. The paint application shall be uniform in thickness and color, smooth in appearance, free of dry over-spray, pinholes, orange peel, sags, or other film defects. Threaded surfaces shall not be painted. All painted areas shall be "touched-up" where paint is removed as a result of handling or assembly at the manufacturer's plant.

3.10.2.1 Primer and intermediate coat. The primer and intermediate coat shall consist of epoxy-polyamide paint conforming to MIL-P-24441. The primer coat shall be green, formula 150 and shall have a minimum 2.5 mil dry film thickness. The intermediate coat shall be white, formula 152 and shall have a minimum 2.5 mil dry film thickness.

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3.10.2.2 Finish coat. All metal surfaces to be painted, shall be given one finish coat of epoxy-polyamide paint conforming to MIL-P-24441. The finish coat shall be haze gray, No. 27, formula 151, and shall have a minimum 3.0 mil dry film thickness.

3.11 Fungus resistance. Electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.12 Labeling of controls. All hand actuated electrical switches, hydraulic control valve levers, hand operated diverters, gages, and areas requiring warning notification of hazards shall be marked by labels. Control panel labels shall be easily readible by the operator when operating the hydraulic power unit while standing at the operator's station. Marking label material shall be securely attached and shall remain legible and durable for the life of the equipment.

3.13 Identification plate. An identification plate will be furnished by the contracting officer for each hydraulic power unit assembly. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix the plate to each hydraulic power unit assembly in a conspicuous place with brass screws or bolts not less that 1/8 inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.14 Workmanship.

3.14.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done accurately. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts may be made, but shall not have a visible notch. All bends of a major character shall be made with controlled means in order to insure uniformity of size, shape, and accuracy. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. Bolt holes and alinement holes shall be accurately punched or drilled and shall have all burrs removed.

3.14.2 Welding. All welding procedures shall be qualified in accordance with section 5 of AWS D1.1. All welding shall conform to the requirements of sections 1 through 7 of AWS D1.1.

3.14.2.1 Welder qualification. All welders shall be qualified for the types of welds and for the positions they are required to use in construction of the fabrications. Qualification shall be on the type of base material and filler utilized in actual construction or on similar alternate materials not requiring operator requalification. Welder qualification shall be performed in accordance with section 5 of AWS D1.1. Copies of the qualification record for each qualified welder shall be kept by the manufacturer or contractor and shall be available to authorized Government inspectors.

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3.14.3 Machine work. Tolerances and gages for metal fits shall conform to the limitations specified herein and on the applicable drawings, or otherwise to the standards of best commercial practice.

3.14.4 Castings and forgings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function. Forgings shall be uniform in quality and condition, and shall be free from tears, cracks, seams, laps, internal ruptures, imbedded scale, segregation, or other defects which may detrimentally affect the suitability for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one complete hydraulic jacking system when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.2 through 4.4.3.4, and the packaging inspection of 4.5.

4.3 Examination. Each production component shall be examined to verify compliance with this specification. Corrections required following failure to meet specified requirements shall receive particular attention for adequacy and suitability. Noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection. Examinations shall include, but not limited to, those listed in the following paragraphs.

4.3.1 Interchangeability examination. Verify fabrications of the same type are interchangeable (see 3.6). Fabrications shall be examined to verify that critical dimensions are within the tolerances specified in the applicable drawings.

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4.3.2 Steel examination. Verify that steel is as specified (see 3.7.1).

4.3.3 Aluminum examination. Verify that aluminum is as specified (see 3.7.2).

4.3.4 Fastener examination. Verify that fasteners conform to proper dimensions, thread characteristics, and specifications (see 3.4 and 3.7.3).

4.3.5 Hydraulic hose assembly and fitting examination. Verify that all hydraulic hose and hydraulic hose fittings are as specified (see 3.7.4). Each hydraulic hose assembly shall be visually examined to reveal cuts, kinks, excessive skiving, or improperly installed fittings.

4.3.6 Welding examination. Visual inspection of welds and weldments shall be performed utilizing the proper mechanical gages, fixtures, and appropriate weld gages. Applicable sections of the AWS D1.1 and the Welding Inspection Manual shall be followed including, but not limited to, verification of the following:

a. Dimensional accuracy of the weldment (including warpage).
b. Conformity to drawing requirements. (This involves determination of whether all required welding has been done, and whether finished welds conform with regard to size and contour.)
c. Acceptability of welds with regard to appearance (including such items as surface roughness, and weld spatter).
d. The presence of unfilled craters, pock marks, undercuts, overlaps, and cracks.
e. Evidence of mishandling from center punch or other inspection markings or excessive grinding.
f. Proper marking of areas requiring weld repair.
g. Subsequent inspection of weld repair.

4.3.7 Cleaning and painting examination. Verify that cleaning procedure, painting, and the time intervals between those processes are as specified (see 3.10 through 3.10.2.2).

4.4 Tests.

4.4.1 Hydraulic hose assembly test. Each hose assembly shall be pressurized, as a minimum, to twice the hose manufacturer's rated working pressure. When the proof pressure is reached, the hose assembly shall be disconnected from the pressure source by a valve. Proof pressure shall be held for not less than 30 seconds and not more than 60 seconds. Any suitable liquid may be used as the pressure medium. Upon completion of the test, the hose assembly shall be cleaned free of the pressure medium. Evidence of slippage or detachment of a fitting, leakage, or rupture shall constitute failure of this test.

4.4.2 Hydraulic power unit assembly test. Each hydraulic power unit assembly shall be operationally tested to verify conformance to 3.8.1 and 3.8.2. Each hydraulic power unit assembly shall be fueled and serviced in accordance with the manufacturer's recommendations preparatory to testing. Each hydraulic power unit assembly shall be operated sufficiently to verify proper functioning of components. Hydraulic power unit assembly controls shall be cycled to reveal any evidence of binding or improper operation. During testing, the hydraulic power unit assembly shall be visually inspected

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to detect fluid leaks. Malfunction of components or controls or fluid leaks or improper assembly functions shall constitute failure of this test and shall be cause for rejection.

4.4.2.1 Hydraulic flow/pressure test. Each hydraulic power unit assembly shall be flow/pressure tested. Hydraulic flow rate and pressure data shall be recorded to verify compliance with 3.8.1 and the applicable NCEL drawings. Compliance shall be verified by measuring the hydraulic oil flow rate and pressure through:

a. Each color-coded group of quick-disconnect fittings on the hydraulic power unit panel simultaneously (see NCEL Drawing No. 76-4-12F).
b. The hydraulic power takeoff (see NCEL Drawing No. 76-4-9F, part No. 43).

4.4.3 Hydraulic jack assembly tests. Each complete hydraulic jack assembly shall be tested to verify conformance to 3.9.2. Each jack shall initially be hydraulically operated sufficiently to verify proper functioning of controls and hydraulically powered components and to reveal any evidence of binding, leaking, or improper operation before performing the following tests.

4.4.3.1 Structural integrity and piston seal leak test. Each jack assembly shall be statically pressure tested to simulate lifting a 50-ton load. Preparatory to statically pressure testing the jack assembly, the piston shall be positioned approximately in the middle of the cylinder's length (piston travel). After positioning the piston, the upper and lower crossarm latches (on each side) shall be locked together by using 1-1/4-inch stud link chain. The lift cylinder shall be pressurized to not less than 3,000 psi to extend the ram. During this test, the piston shall not contact the cylinder head and the piston shall remain approximately near the middle of the cylinder. Piston packing shall hold the load, 50 tons, without any evidence of internal or external leakage for not less than 6 minutes. Malfunction of components or permanent deformation of components or hydraulic fluid leakage shall constitute failure of this test and shall be cause for rejection.

4.4.3.2 Gland seal leak test. The hydraulic jack cylinder shall be pressurized to not less than (3,000 psi) after fully retracting the piston and rod (to the bottom of the jack cylinder). Pressure shall be held while visually inspecting the outside area of the gland seal. Evidence of leaking hydraulic fluid or seal assembly movement shall constitute failure of this test and shall be cause for rejection.

4.4.3.3 Cylinder structural integrity and piston seal leak test. The hydraulic jack cylinder shall be pressurized to not less than (3,000 psi) to fully extend the piston and rod (to the top of the jack cylinder). Pressure shall be held while visually inspecting the outside area of the jack cylinder. Malfunction of components or permanent deformation of components or hydraulic fluid leakage shall constitute failure of this test and shall be cause for rejection.

4.4.3.4 Control box assembly flow/pressure test. Each control box assembly shall be flow/pressure tested to verify conformance to 3.9.4 and compliance with NCEL Drawing Nos. 76-4-4F and 76-4-13F. Hydraulic flow rate and pressure

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readings shall be recorded to verify compliance with applicable NCEL drawings. All components shall be visually inspected for leakage during the pressure test. Evidence of leaking hydraulic fluid or improper assembly functions shall constitute failure of this test and shall be cause for rejection.

4.4.4 System test. The complete jacking system shall be tested as an assembly. Four 50-ton hydraulic jacks shall be hydraulically coupled, via the appropriate hose bundles, to the appropriate color coded quick disconnect fittings on the hydraulic power unit. Each control valve which operates jack functions shall be cyclically operated not less than twenty five times to verify unit operability and proper control function. Evidence of malfunctions, mislabeling of controls, or internal/external leakage shall constitute failure of this test and shall be cause for rejection.

4.5 Packaging inspection. The preservation, packing, and marking of the complete hydraulic jacking system shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications. Grease conforming to MIL-G-18458 shall be substituted for type P-11 preservative.

5.1.1.2 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed, and mating parts on the equipment, shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved as follows:

5.1.1.4.1 Unfinished (not machined) surfaces. Unfinished exterior metal surfaces shall be coated with type P-1 preservative.

5.1.1.4.2 Machined surfaces. Exposed machined surfaces shall be coated with type P-6 or P-11 preservative and wrapped or covered, as applicable, with barrier material conforming to MIL-B-121, type I, grade A, class 2. The material shall be secured in place with waterproof tape.

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5.1.1.5 Engines. Engines, engine components, and accessories shall be preserved in accordance with the level A requirements of MIL-E-10062, type I, method I.

5.1.1.6 Hydraulic systems. The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier-material conforming to MIL-B-121, type I, grade A, class 2, extending the wraps approximately 2 inches onto the ram cylinders. The wraps shall be secured in place with waterproof tape. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. A weatherproof tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Markings shall be applied to the tag with a waterproof material.

5.1.1.7 Hose. The hose shall be coiled to a diameter to prevent deformation or kinking and secured.

5.1.1.8 Drive belts and pulleys. Drive belts shall be removed or released from tension. Removed belts shall be preserved method IC-1 or IC-3. Unpainted surfaces shall be coated with primer conforming to TT-P-664 or MIL-P-46093. A weatherproof tag shall be attached in a conspicuous location indicating: "Belts have been (removed or released from tension). (Install or tension) prior to operation." Markings shall be applied to the tags with a waterproof material.

5.1.1.9 Loose items. Items not secured within the shelter shall be individually preserved, method I or III, as applicable, and consolidated packaged as specified in 5.1.1.10.

5.1.1.10 Consolidation. Disassembled items, loose items, and packaged items not secured within the shelter for each complete hydraulic jacking system shall be consolidated in containers conforming to PPP-B-601, domestic type, or PPP-B-621, class 1. Contents shall be cushioned, blocked, and braced to prevent movement.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing.

5.2.1 Level A. Components of the hydraulic jacking system shall be arranged within the equipment shelter and secured in such a manner to prevent movement. Consolidation containers shall also be secured in such a manner to prevent movement. The shelter doors shall be secured by lock or seal.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The equipment covered by this specification is to be used in conjunction with miscellaneous lift hardware, not covered herein, for raising and lowering pontoon causeway sections in surf zones and beach areas.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When specification and standards shall be other than as specified (see 2.1.1).
c. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.5).
d. Quantity of components required (see 3.9).
e. When equipment shelters shall be furnished by the Government (see 3.9.9 and 6.3).
f. Level of preservation required (see 5.1).

6.3 Government-furnished property. The contracting officer should arrange to furnish equipment shelters, which conform to MIL-S-28633, when required (see 3.9.9 and 6.2 (e)).

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Preparing activity:

Navy - YD

Project No. 1945-N045

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-29214(YD)	HYDRAULIC JACK SYSTEM, ELEVATED CAUSEWAY: HYDRAULIC COMPONENTS AND SHELTER PROVISIONS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426

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